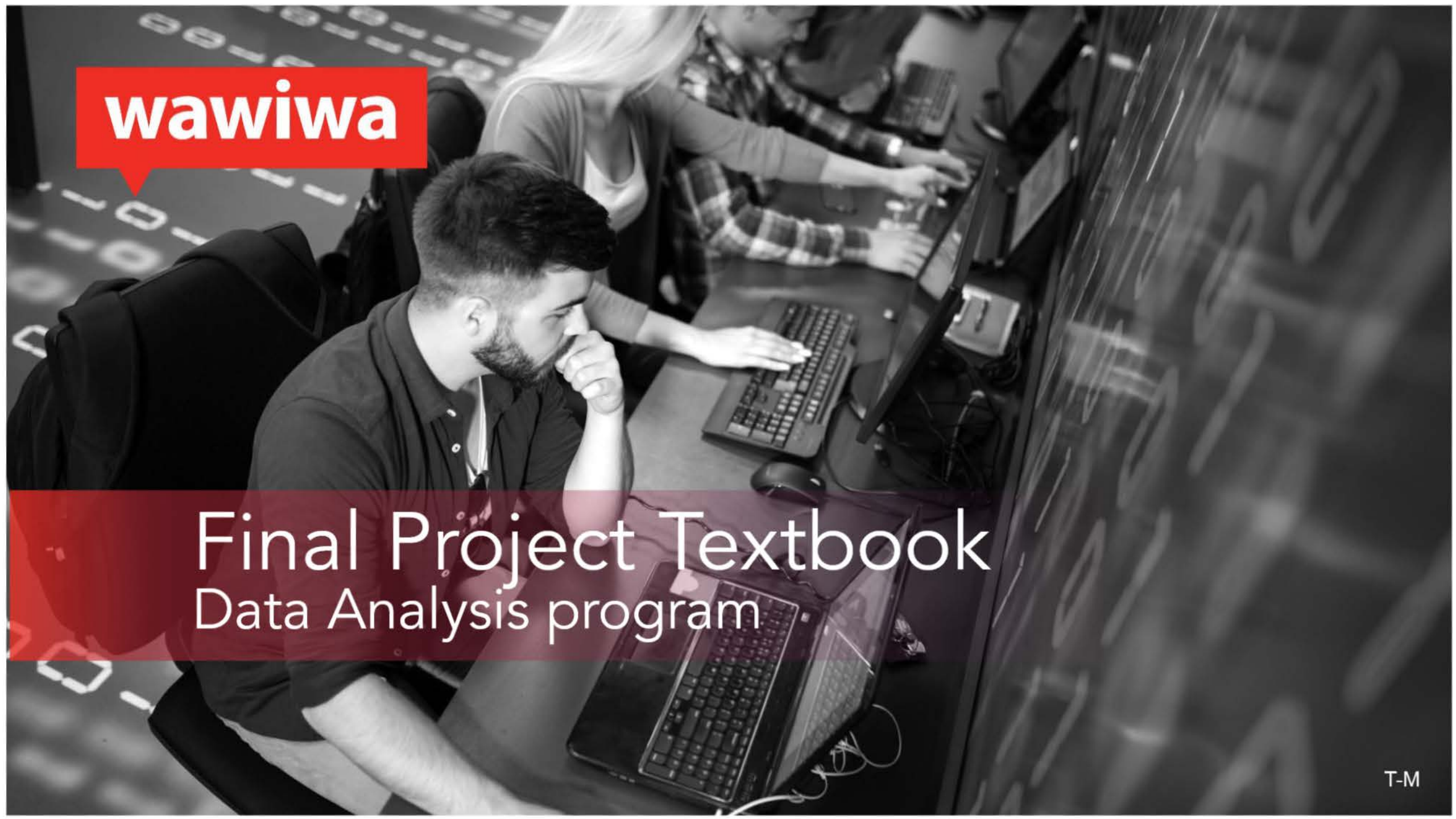




**wawiwa**

# Final Project Textbook

## Data Analysis program

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# General Instructions for the Final Project

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## Lesson 0

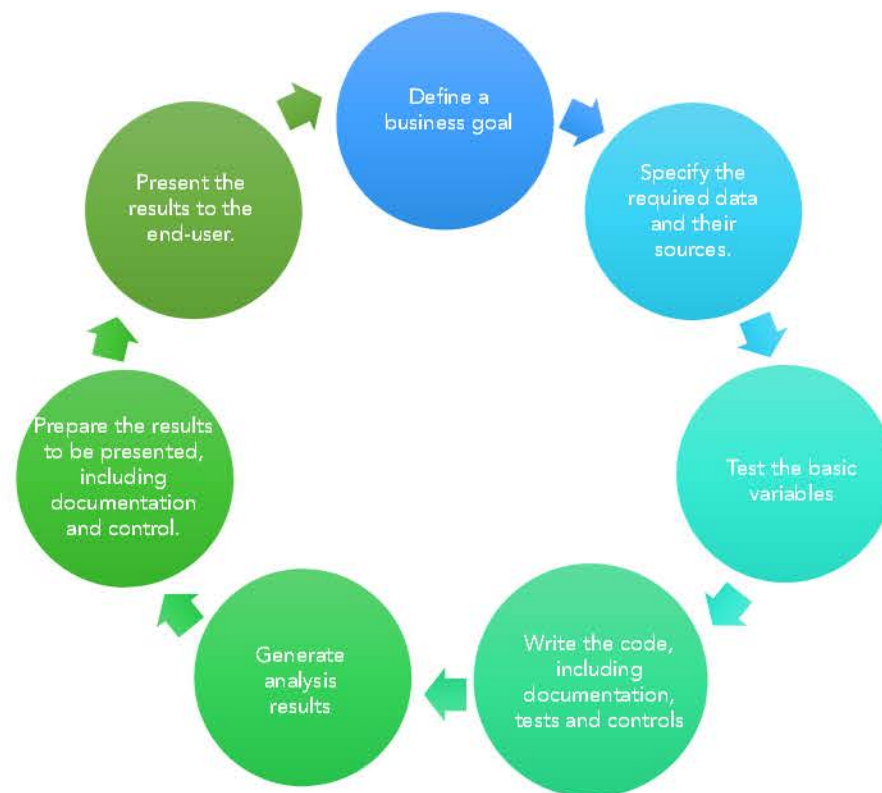


# Overview of the Final Project

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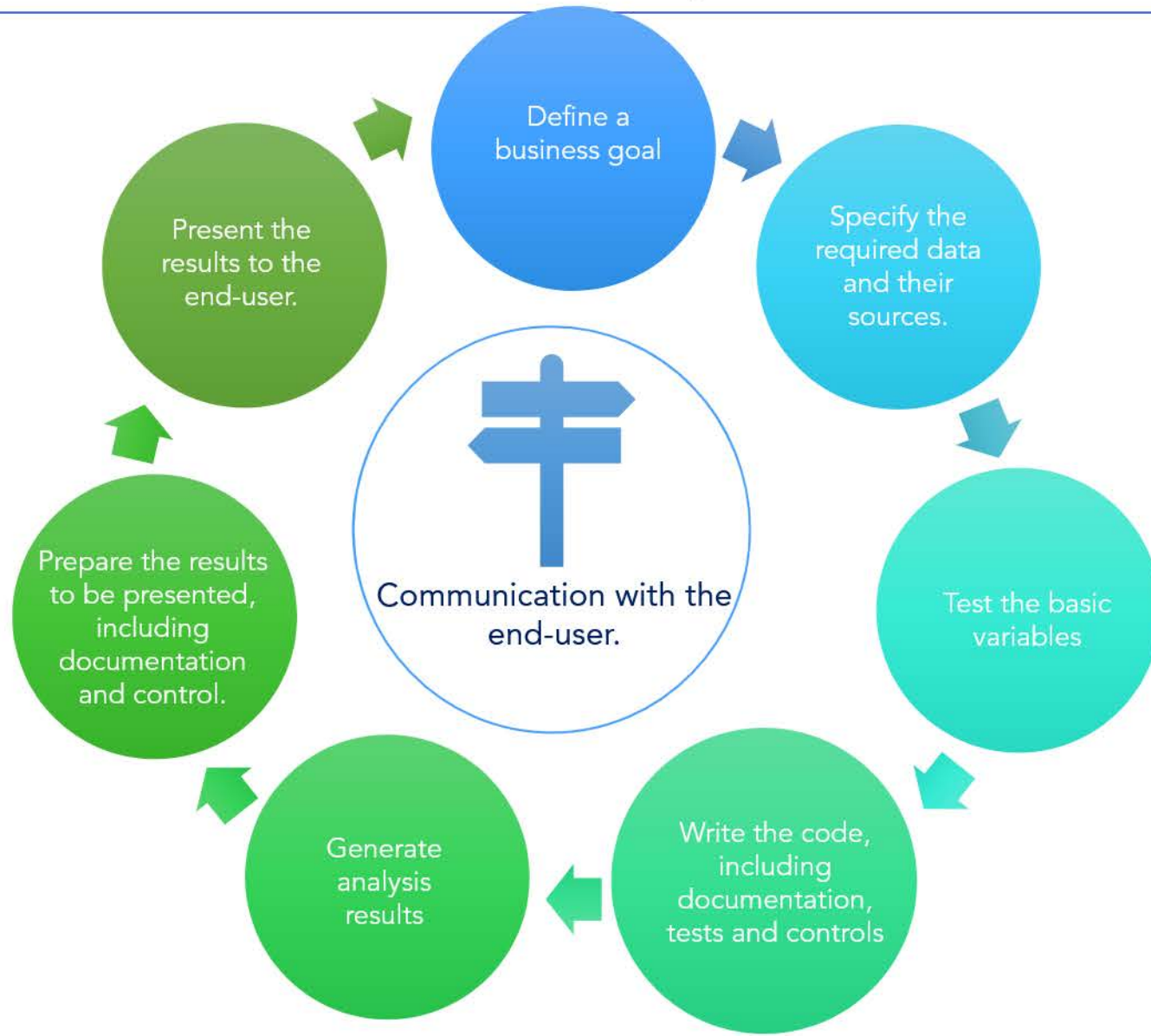
# Purpose

The Final Project for the Data Analyst Course simulates the work of an analyst from the very beginning right up to submitting the conclusions to the business client, working according to the stages of the analyst's work cycle.





# Data Analysis with the Analyst's Work Cycle



# Overview of the Stages in the Final Project

---

1. Form groups according to the lecturer's instructions.
2. Each group receives a data set to analyze.
3. Perform the analysis according to the steps in the analyst's work cycle.
4. Each group must select the tool or tools they will use to analyze the data.  
(If you choose Excel, you must select at least one additional tool.)
5. Present the results during the course summary panel.

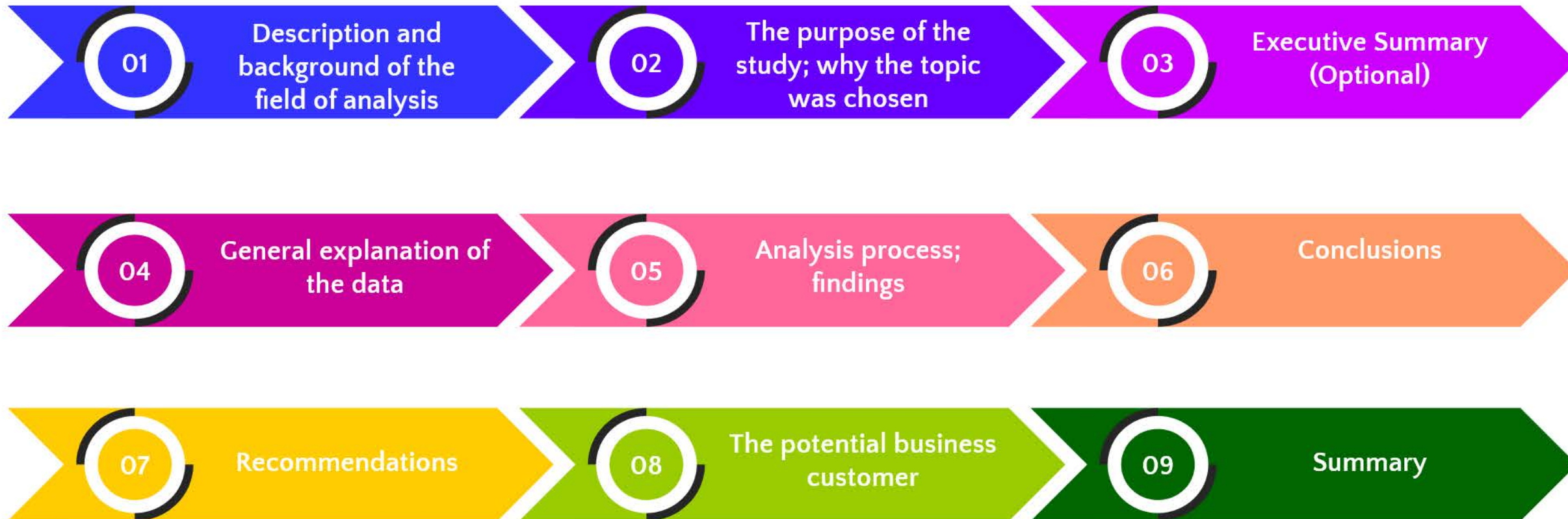
# The Course Summary Panel

---

**Each group will have up to 15 minutes to  
present their project.**

**Please be careful to remain within the  
allotted timeframe.**

# Suggested Presentation Layout





# Outcomes of the Final Project

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The outcomes submitted will include:

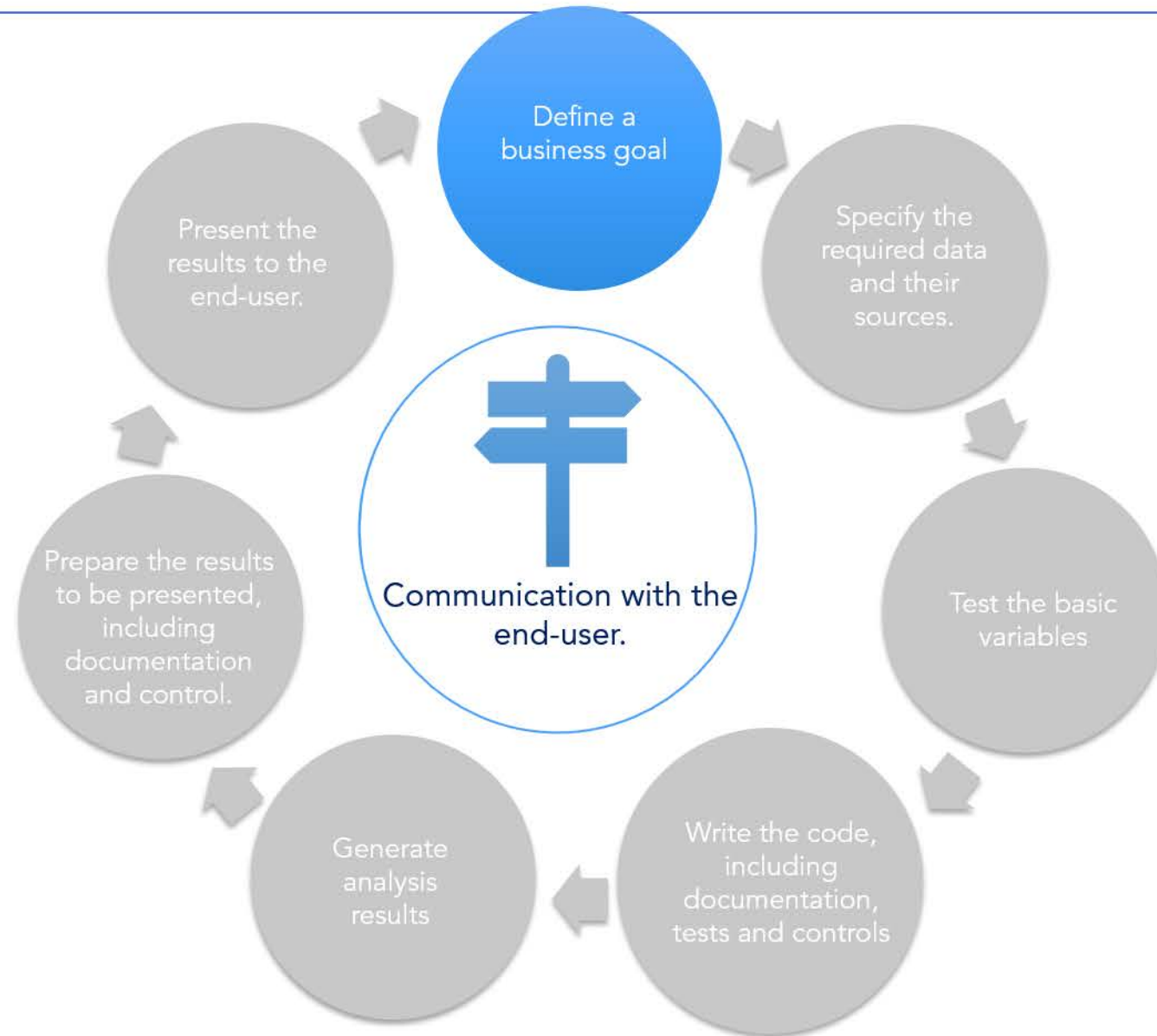
- A visual display (presentation / dashboard / other) of the process and the results
- External documentation
- All of the analyses in SQL, Excel, Power BI and Python



## Overview of the Initial Stages of the Analysis Process

---

# Overview of the Work Stages in the Final Project

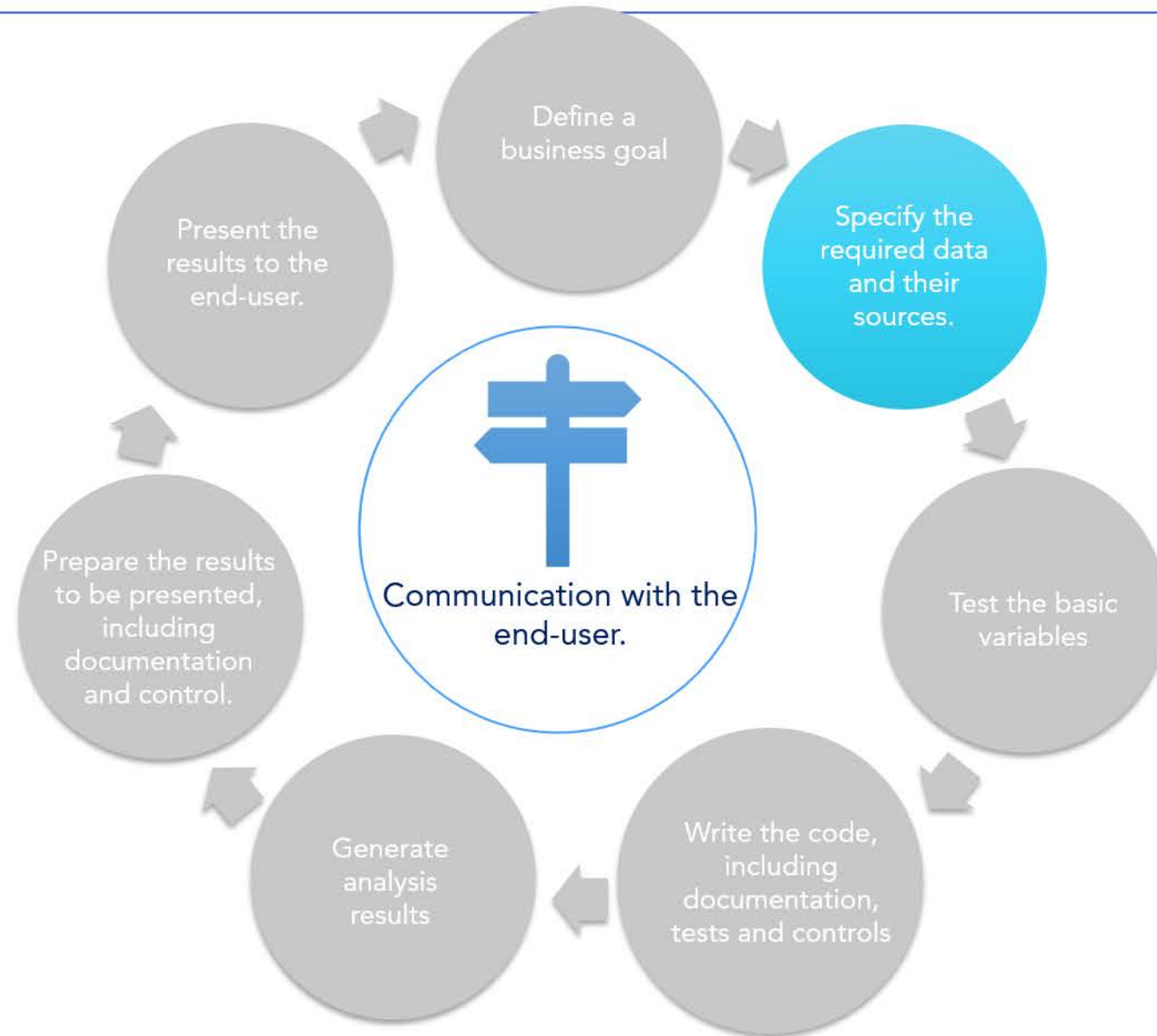


# Define a Business Research Question

---

- ❑ Examine the data set and understand the meaning of the columns and the relationships between the columns.
- ❑ Formulate a business research question, taking into consideration the following points:
  - What is the purpose of the analysis?
  - Who is the potential business customer for this type of data analysis?
  - What value can the potential business customer derive from the results of the analysis?

# Overview of the Work Stages in the Final Project





# Define the Required Variables and their Sources

---

- ❑ Review the data set and define which columns are relevant to the analysis (data panel).

There can be columns of several types:

- Columns that appear as they are in the dataset
- Columns calculated from the data set
- Columns added from external information source



## Task Required for the Next Project Status Meeting

---

# For the Next Project Status Meeting ...

---

- ☐ Form groups.
- ☐ Familiarize yourselves with and understand the data source.  
(If there are questions, search online for answers.)
- ☐ Formulate a research question.
- ☐ Prepare a broad definition of the variables and sources required in order to answer the research question.

# For the Next Project Status Meeting ...

---

❑ Test the data that is relevant for the analysis for:

- ✓ Accuracy
- ✓ Feasibility

Reminder: The test set depends on the type of variable:

- Categorical variable
- Continuous variable

❑ If there are incorrect / missing / exceptional data, consider how they should be handled.  
(In the next project status lesson, we will address the subject more thoroughly.)

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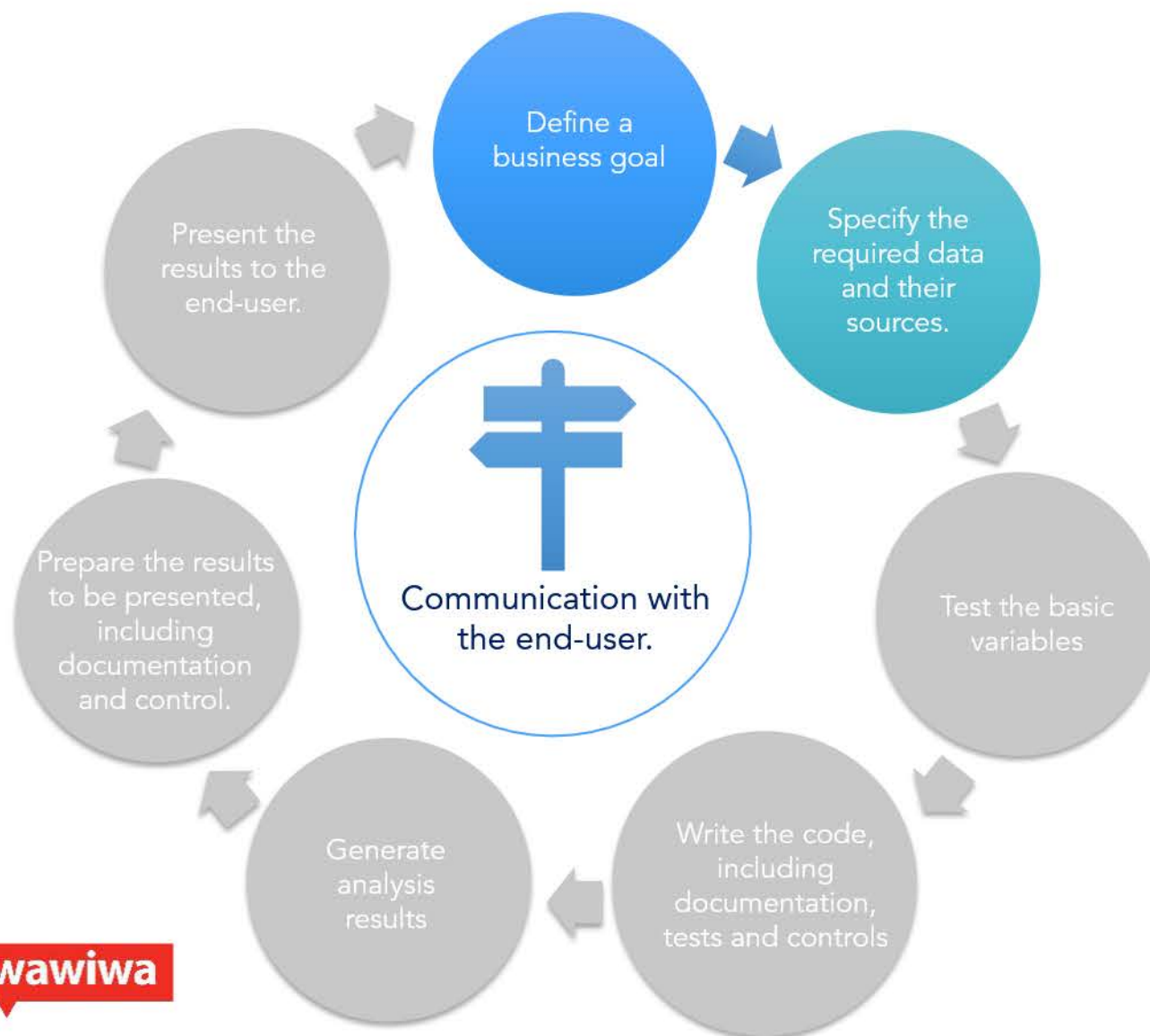
# Handling Missing and Abnormal Data

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## Final Project - Status Meeting 1



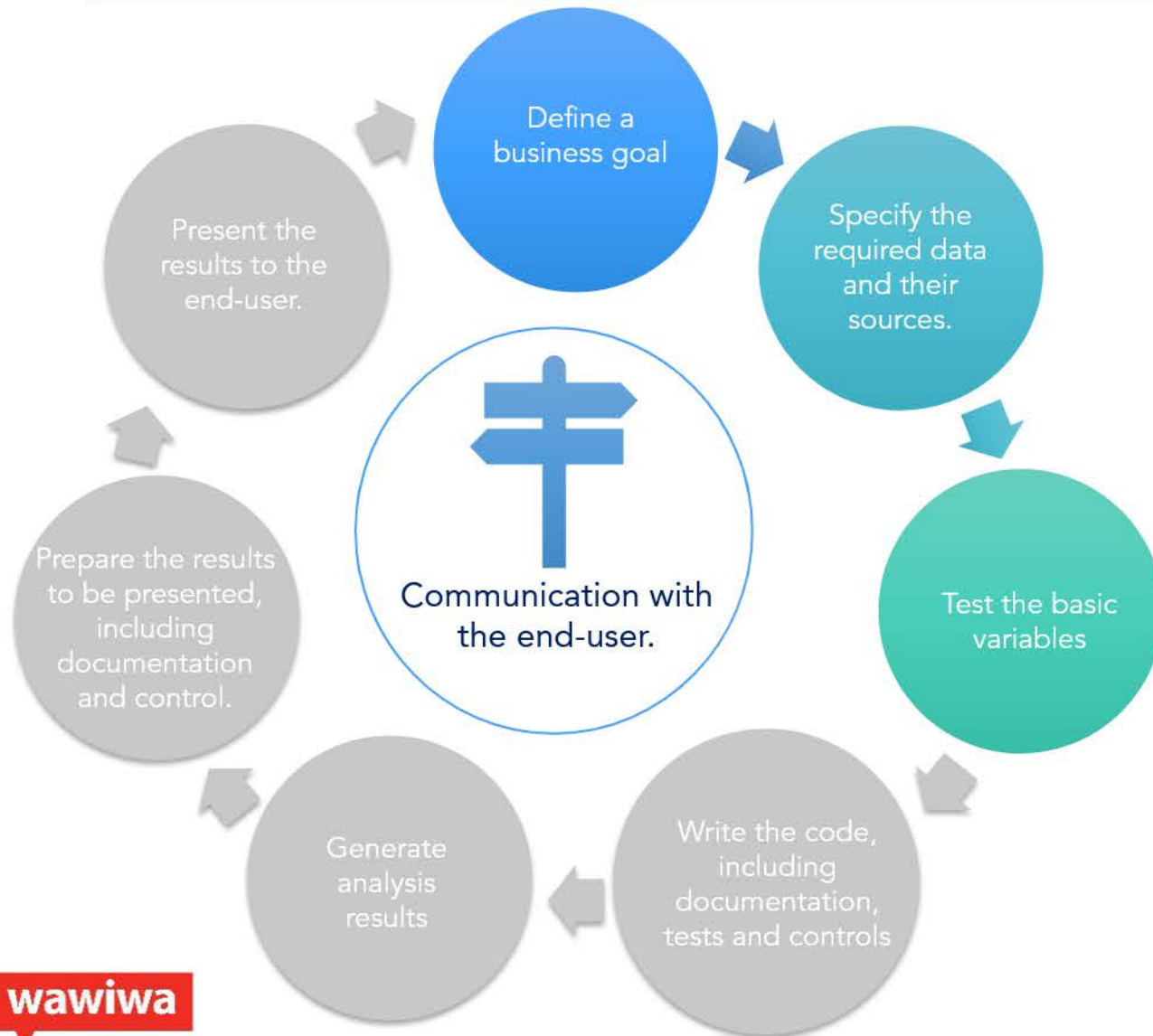
# Data Analysis with the Analyst's Work Cycle



In the previous stages we dealt with:

1. Defining the business goal
2. Specifying the required variables

# Data Analysis with the Analyst's Work Cycle



Today we will proceed to the next stages:

1. Checking the raw data
2. Writing the code, including documentation



# Checking the Raw Data

---

# Types of Variables

---

## ❑ Categorical Variable

The values are used only as labels.

For example: customer type, city code, color, category code, etc.

## ❑ Continuous Variable

The values are ranged on an infinite scale of numbers, including decimals.

For example: salary, sale price, weight, quantity, discount percentage/amount, distance, etc.

# Abnormal Data Types

---

- ☐ Extreme values (in a continuous variable)
- ☐ Missing values (NULL)
- ☐ Incorrect values (spelling errors, illogical values or values outside the logical range)

Analytical comprehension is required to identify an abnormal value.

# Locating Abnormal Values

---

For each variable:

- ☐ Identify the nature and meaning of the variable and its values.
- ☐ Recognize the variable type (Continuous or Categorical).
- ☐ Search for abnormal values.



# Handling Abnormal Values

---

Possible actions for abnormal values:

- ☐ Remove the variable (field or column).
- ☐ Delete the entry (data row).
- ☐ Replace it with an alternate value (percentage, average, median, frequent, exception model, etc.)

# Handling Abnormal Values

---

How to handle abnormal data:

- Create another column in the table (ColNameFix) for the values after handling. In this way, the original values are retained for validation and backup.
- Add an explanation in the external documentation:
  - What data was handled
  - How it was handled
  - Why it was handled in this way / Why another method was not used.

# Handling Abnormal Values

---

Remember that the treatment must be  
tailored to the business rationale and  
.coordinated with the business entity

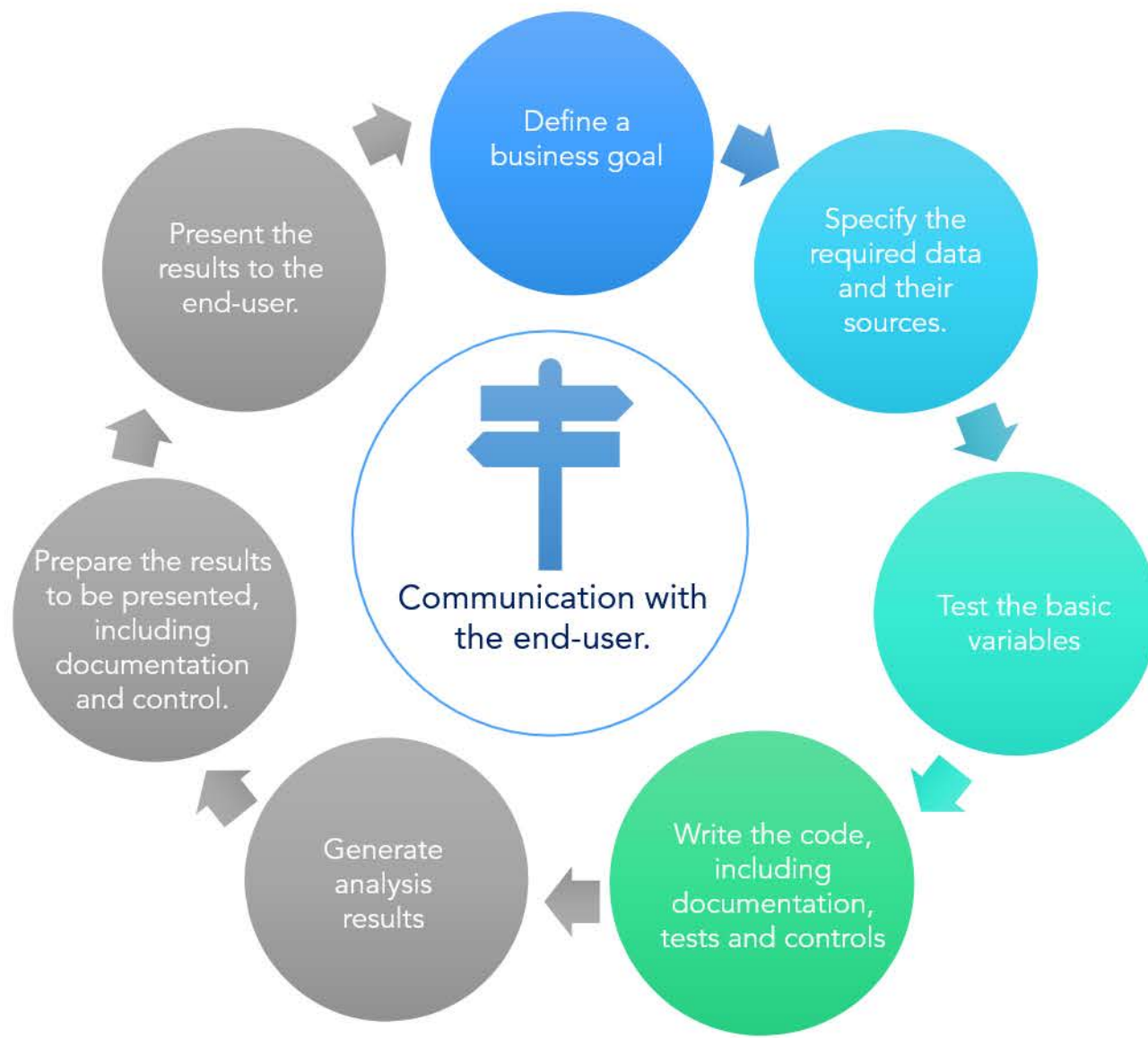




## Writing the Code Including Documentation, Controls & Testing

---

# Data Analysis with the Analyst's Work Cycle



# Preparing to Write the Code

---

After having understood the nature of the fields, checked their correctness and decided which fields to use to analyze the data, it is possible to proceed to preparation for writing the code.



# Preparing to Write the Code

---

1

To work in an orderly manner, first define in detail  
**what is to be tested.**

In other words, formulate sub-questions to the main business question, so that answering these sub-questions constitutes, in itself, the analysis process.

# Writing the Code

---

2

After having defined the questions,  
solve them using one or a combination of the technical  
tools studied:

MSSQL, Excel, Power BI, and Python

# Checking the Code

---

3

Checking at each stage that the code is correct and returns logical, accurate results is important, and even critical.

If the calculations are incorrect,  
the whole analysis is incorrect!

Good Luck! 😊

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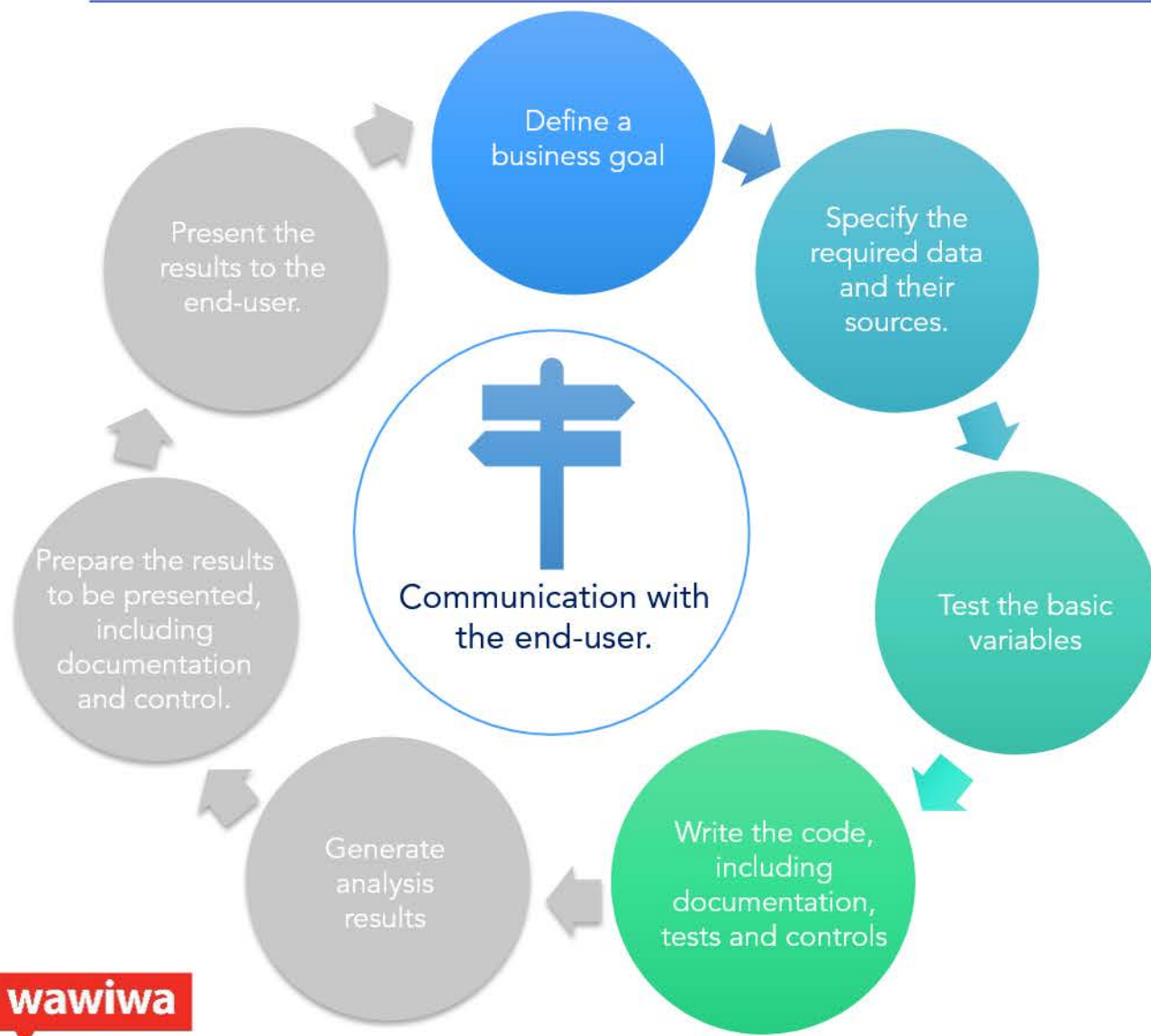
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# The Analysis

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## Final Project - Status Meeting 2

# Data Analysis with the Analyst's Work Cycle

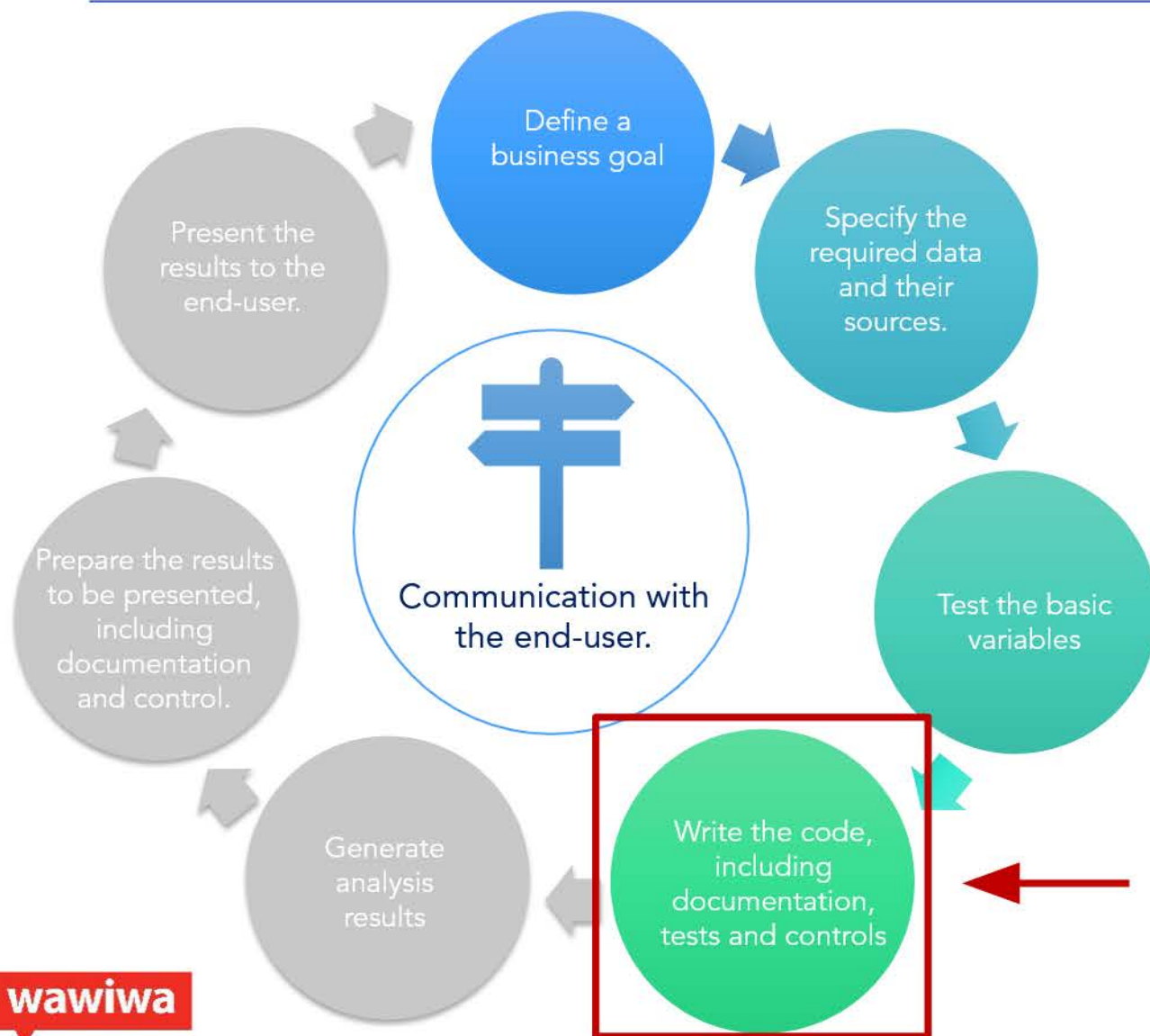


In the previous stages we dealt with:

1. Defining the business goal
2. Specifying the required variables
3. Checking the raw data
4. Writing the code, documentation and controls



# Data Analysis with the Analyst's Work Cycle



Today we will continue working on the analysis:

- Writing the code, documentation and controls



# Preparing to Write the Code

---

In the previous session we took the business question and broke it down into subsections.

Investigation and analysis of the subsections will provide the information from which to formulate the answer to the business question of the final project.

# Analysis of the Data

---

1

Specify the properties of each section, **including details**:

- What is the overall goal?
- What data must be displayed?
- Should the information be filtered? (By time period? Ages?)
- Should the information be grouped? (By categories? Areas?)
- Which sorting is most suitable for the display?

# Planning the Presentation of the Results

---

2 While analyzing the data, **consider how to present the answer** to the subsections in a way that will allow the viewer to understand clearly the result you want to convey.

**.Prepare the analysis to suit the plan**

# Planning the Presentation of the Results

---

2

Examples of presentations:

- Comparative chart
- Heat distribution map
- Dashboard
- A chart with explanatory comments

# Checking the Code

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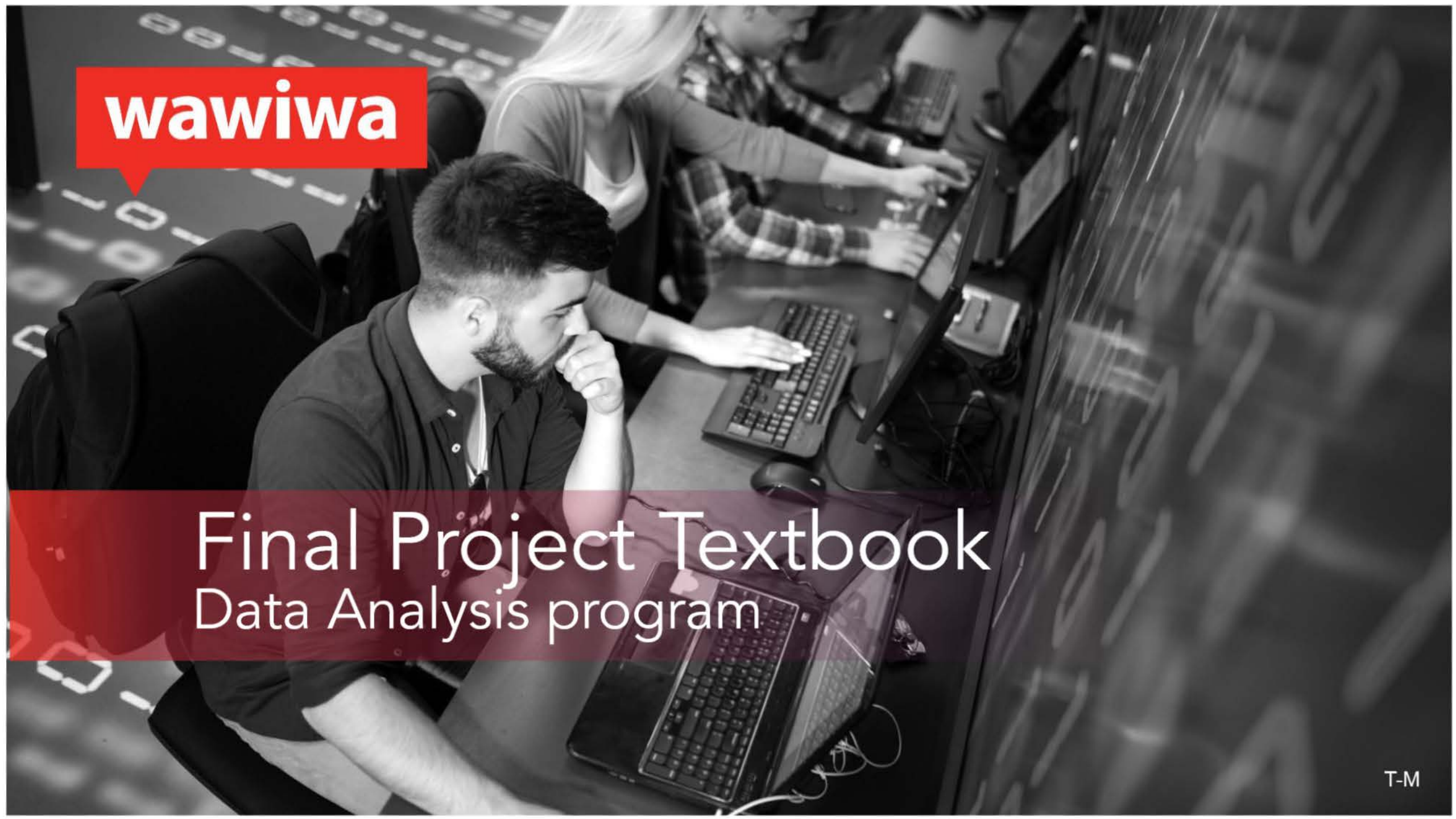
3

We cannot stress enough that it is of the utmost importance to

Check, recheck and check again  
the calculations and results of the analysis.

Good Luck! 😊





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# Final Project Textbook

## Data Analysis program



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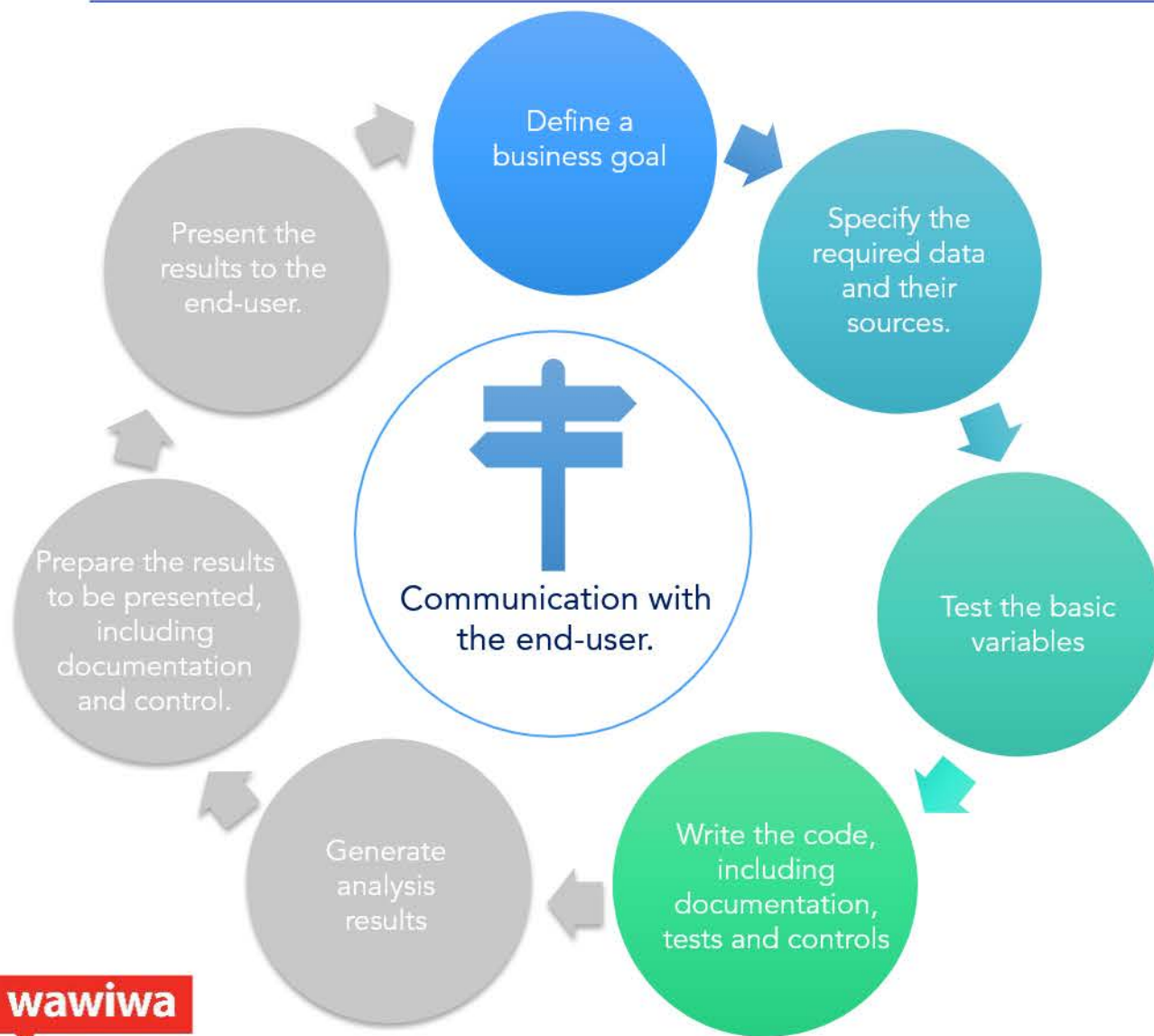
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Preparing the results

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Final Project - Status Meeting 3

# Data Analysis with the Analyst's Work Cycle



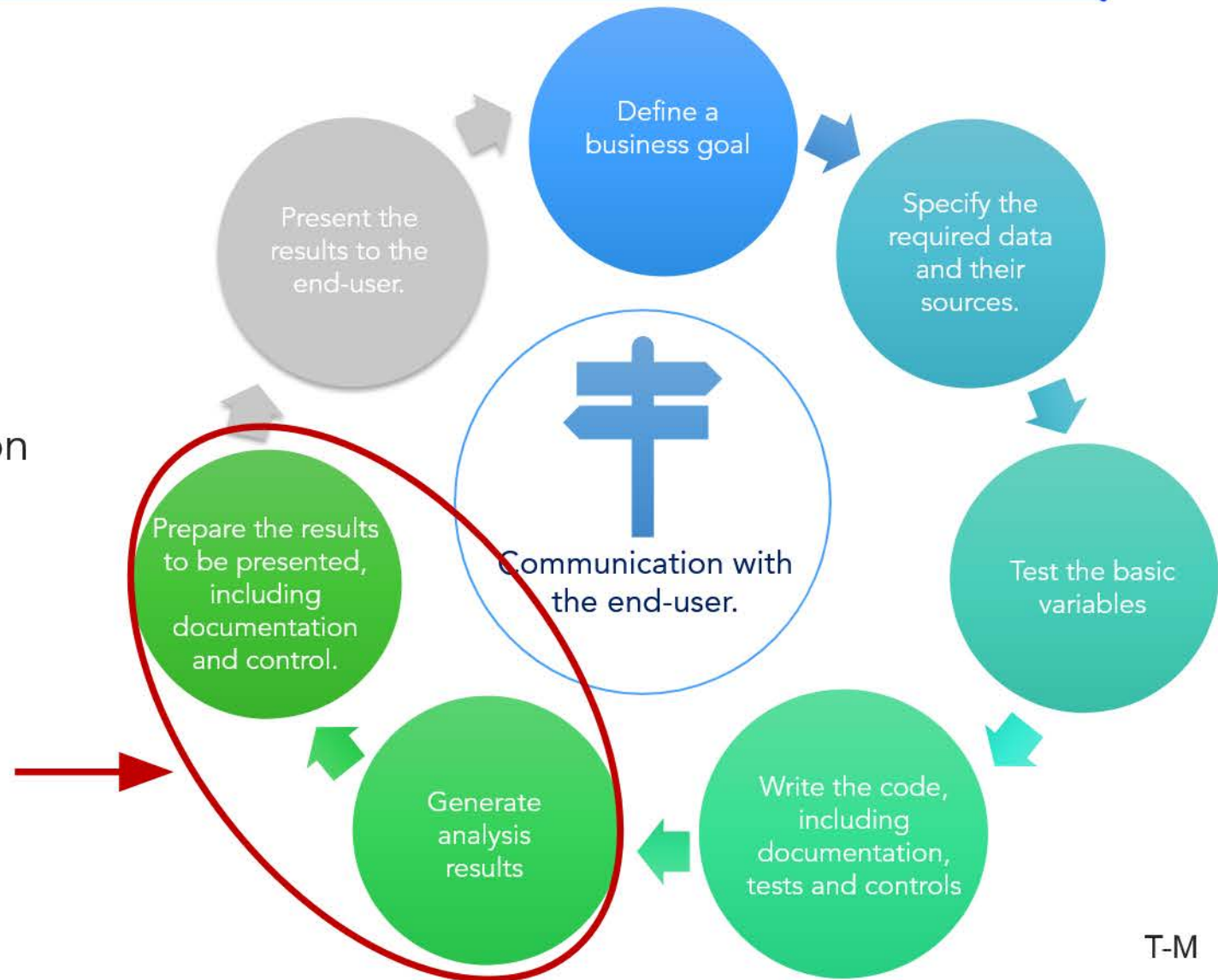
In the previous stages we dealt with:

1. Defining the business goal
2. Specifying the required variables
3. Checking the raw data
4. Writing the code, documentation and controls

# Data Analysis with the Analyst's Work Cycle

Today we will continue working on the analysis:

1. Generating the analysis results
2. Preparing the results for presentation



# Generating the Analysis Results

---

After the results of the analyzed subsections have  
been obtained,

prepare the results for visual presentation

in a slideshow or any other type of presentation.

# Conclusions

---

For each sub-query, prepare 1-2 slides in which one of the following is presented visually :

- a chart
- a screenshot of the dashboard
- a switch to the dashboard during the presentation to show the use of the dashboard real-time and the results
- Any other form of visual presentation that displays the outcome



# Conclusions

---

To conclude the presentation of the results,

recap the main, important conclusions

(up to 5 or 6) in one summary slide.





# Recommendations

---

After having presented the conclusions,

**recommend changes to be made**

.based on the conclusions obtained from the data analysis

(.In addition, points for further analysis may be suggested)

# Important Points

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How to design the presentation –

:Pay attention to the following points

- The text in the slides must be concise
- In each slide of a chart, it is best to include a sentence that summarizes the conclusions
- Highlight the parts of the chart that the viewer should focus on
- Format the presentation in an orderly, clear, easy-to-understand layout



continued...

# Important Points

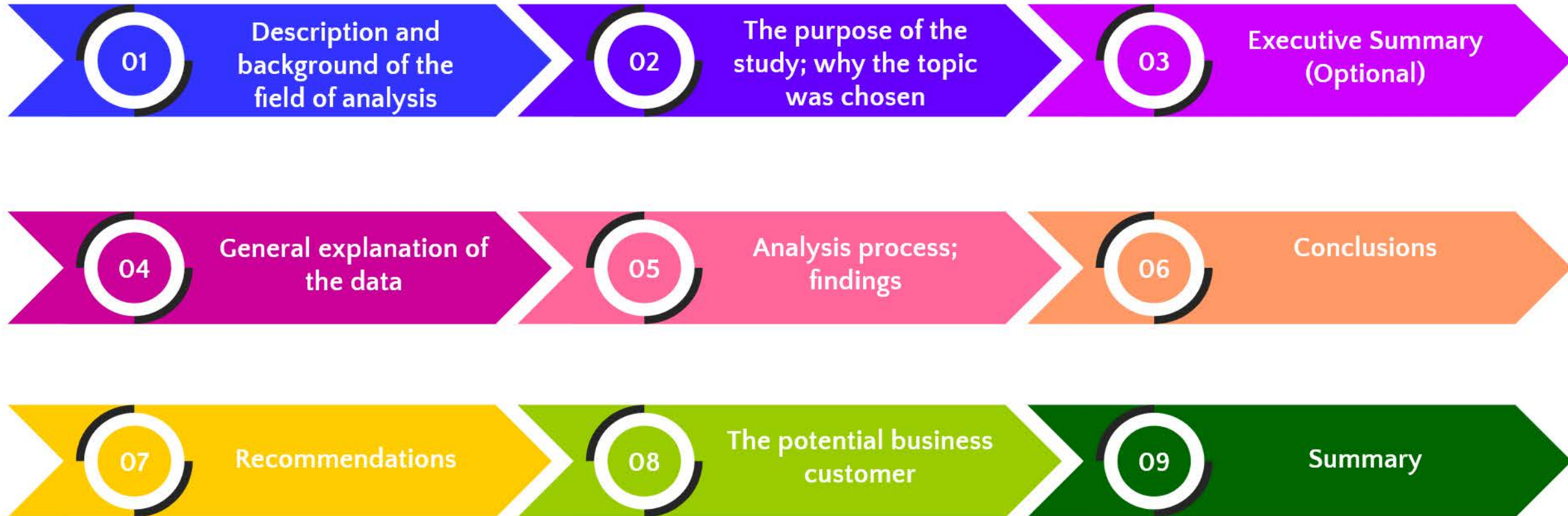
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continued...

- Pay attention to the logical order of the slides.  
The presentation should tell us the story of the data analysis.
- Be sure to give some background about the field of the data, and to suggest the potential business customer.  
(Who would pay for it and earn from it? How could they profit from the analysis and the conclusions?)



# Suggested Presentation Layout



# Final Project Grading

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:The grade of the final project will be determined on the basis of the following criteria

- The business goal
- The process of building the panel and using the data
- Quality of the presentation of the data analysis
- Presentation format and design
- Presentation of conclusions and recommendations
- Overall impression







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That's it for today,  
see you next time!